XIAMEN UNIVERSITY MALAYSIA

Online Quiz 1

Course Code:	COM108		
Course Name:	Engineering Maths (I)		
Question Paper Setter:	Liu Mei Feng		
Academic Session:	2021/09	Question Paper:	A ☐ B ☐
Total No. of Pages:	1	Time Allocated:	24 hours
Additional Materials:	-		
Apparatus Allowed:	-		

INSTRUCTIONS TO CANDIDATES

1. There are four questions in this paper. Please do ALL questions.

2. Communication between candidates in any means is forbidden. Answers must be entirely individual candidate's independent effort. If you are found sharing your solutions with other candidates, or suspected of doing so, you would be penalized accordingly.

(Student ID: _____ Full Name: _____)

Question 1 (15 marks)

Given a function

$$f(x) = \frac{1 - e^{-x}}{1 + e^{-x}}$$

(a) Sketch the graph of the function. [3 marks]

(b) State the domain and the range of the function. [2 marks]

(c) Test if the function is one-to-one. [2 marks]

(d) Find the inverse function $f^{-1}(x)$ if any. [6 marks]

(e) What is the domain and range of $f^{-1}(x)$, if it exists? [2 marks]

Question 2 (10 marks)

Given a function

$$g(x) = \begin{cases} x - 2 & \text{if } x \leq 0 \\ \sqrt{x} & \text{if } x > 0 \end{cases}$$

(a) Sketch the graph of the function. [2 marks]

(b) Test if the function is one-to-one. [2 marks]

(c) Determine whether the inverse function exists or not. [2 marks]

(d) Find the inverse function $f^{-1}(x)$ if any. [4 marks]

Question 3 (15 marks)

Let the function be defined as

$$h(x) = \frac{4x^2 + kx + 7k - 6}{2x^2 - 5x - 3}$$

where k is an undermined real constant.

(a) Find the domain of the function. [2 marks]

(b) If $\lim_{x \to 3} h(x)$ exists, find the value of the corresponding k. [2 marks]

(c) Find the value of $\lim_{x \to 3} h(x)$ in (b). [4 marks]

(d) If $\lim_{x \to -1/2} h(x)$ exists, find the value of the corresponding k. [3 marks]

(e) Find the value if $\lim_{x \to -1/2} h(x)$ in (d). [4 marks]

Question 4 (10 marks)

Evaluate the following limit or show it does not exist.

(a) $\lim_{x \to 2} \frac{\sqrt{x+2}}{x-4}$

(b) $\lim_{x \to 4} \frac{2x^3 - 128}{\sqrt{x} - 2}$

- END OF PAPER -